Botao Zhang

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Education

Fudan University *Shanghai, China*
B.S. in Mathematics, School of Mathematical Sciences Sept 2023 – Jul 2027 (expected)

Shanghai High School *Shanghai, China*
Sept 2020 – Jun 2023

Research Experience

Benna Lab, University of California, San Diego *San Diego, CA, USA*
Visiting Student · Advisor: Prof. Marcus Benna Jan 2026 – Present

- Developed a biologically plausible neural network that decomposes patterns into features, inspired by SNMF algorithm's derivation from Cengiz Pehlevan's lab. This decomposition allows patterns to be represented sparsely as combinations of features, enabling efficient storage in a classical Hopfield network.

Laboratory of Cognitive Computational Neuroscience and Neuroimaging
Shanghai Jiao Tong University *Shanghai, China*
Visiting Student · Advisor: Prof. Ruyuan Zhang Oct 2023 – Dec 2025

- Investigated arbitration between model-free and model-based control in decision-making. Building on the reliability-based arbitration framework of Lee et al. (2014), developed and tested a simpler, equally intuitive arbitration model to explain behavioral data.

Technical Skills

Programming: Python, R, LaTeX
Languages: Mandarin Chinese (native), English (fluent)

Awards

First Prize, Chinese National High School Mathematics Olympiad, Shanghai 2021, 2022